

Question 5 — Advanced Multi-State Traffic Controller

Register Configuration

The traffic lights are connected to PORTB:

- **Red LED** → **PB0**
- **Yellow LED** → **PB1**
- **Green LED** → **PB2**
- **Pedestrian LED** → **PB3**

These pins are configured as outputs using DDRB, and the LEDs are controlled directly with bit masking.

The pedestrian request button is connected to **INT0 / D2**. The external interrupt is configured using the EICRA and EIMSK registers. When the button is pressed, the interrupt sets a **pedRequest** flag.

Timing Calculations

The controller uses the following state times:

- **GREEN:** 6000 ms
- **YELLOW:** 2500 ms
- **RED:** 5000 ms
- **PED CROSS:** 4000 ms

If the pedestrian button is pressed during GREEN before 3000 ms has passed, the system waits until 3000 ms and then moves to YELLOW. If the request happens after 3000 ms, it moves to YELLOW immediately.

During the pedestrian crossing state, the pedestrian LED flashes at 2.5 Hz using non-blocking timing.

The LCD is updated every 200 ms to show the remaining time for the current state.

System Behavior

The system normally moves through:

GREEN → YELLOW → RED

When a pedestrian request is received, the controller safely changes the traffic sequence before entering the PED CROSS state.

During PED CROSS:

- The red traffic LED remains ON.
- The pedestrian LED flashes.
- New pedestrian requests are ignored until the crossing state is finished.

The LCD shows the current traffic state on the first row and the remaining state time on the second row.

All timing and state changes are handled using `millis()` and direct register operations without blocking the main program.